Truthful Verified Credit: Rewarding Verified Value, Not Reliance, Among Learning Agents

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Abstract

We study the problem of how to credit each self-interested agent for its own contribution to a shared return that incentivises them to truthfully report to maximise social welfare. We propose truthful verified credit (TVC), which rewards only verified value, adds a bonus for every checked claim, and fines audited false claims, and can be used as a supplement of an existing credit mechanism. In a multi-agent bandit, TVC cuts exaggeration from over 90% of reports to below 1% at welfare comparable to popularity credit, and combined with credit for exploration it keeps reports honest at the highest welfare. In multi-robot pathfinding on POGEMA, it cuts the hype rate of self-interested learners from 99% to 0.2% and triples lifelong throughput, and lying stays below 5% with DQN and PPO learners.

Keywords: credit assignment, incentive design, truthful reporting, multi-agent learning, mechanism design

1 Introduction

In a review of FDA records for twelve antidepressants, 94% of the trials in the published literature were positive, against 51% in the FDA's own analysis, and the published effect sizes were 32% larger overall (Turner et al., 2008). The pattern is not confined to medicine: in a set of peer-reviewed social-science experiments, strong results were 60 percentage points more likely to be written up than null results (Franco et al., 2014). Selection also acts inside a study. Comparing trial protocols with their publications, 40–62% of trials had at least one primary outcome changed, introduced or omitted, and statistically significant outcomes were more likely to be fully reported (Dwan et al., 2008). This is publication bias and reporting bias, the file drawer problem of Rosenthal (1979), and it needs no fraud: small studies, flexible analyses and competing interests already make many published findings likely to be false (Ioannidis, 2005), and incentives that reward publication can select for weak methods without anyone cheating deliberately (Smaldino and McElreath, 2016). Authors respond to the credit they receive, and a literature that credits positive, attention-getting results teaches them to report only those.

Behind these figures lies the intersection of credit assignment problem and incentive mechanism design. The problem is how to credit each self-interested agent for its own contribution to a shared return that incentivises them to truthfully report to maximise social welfare. Agents that search on behalf of a group bear the cost of exploring, but what they find can save everyone else the same work. Therefore, a group gains only when its members explore and then report truthfully what they find. Models of the division of cognitive labor show that the reward for discovery shapes which problems a community

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chooses to pursue (Kitcher, 1990; Strevens, 2003), and the same dependence appears when a planner decides what to disclose to agents whose own experiments create the information others rely on (Kremer et al., 2014; Mansour et al., 2015). A growing family of methods now rewards agents directly for their effect on others: social-influence rewards credit an agent for changing other agents' actions (Jaques et al., 2019), and learned incentive functions let agents reward one another to shape behavior (Yang et al., 2020). How credit rewards exploration also decides whether members report what they find honestly.Subsequently, they learn to hack the mechanism by deception (Pan et al., 2022; Wang et al., 2026).

Theory explains why such failures persist. Conflicts of interest limit how much information survives strategic communication (Crawford and Sobel, 1982), receivers who take messages at face value are deceived by inflated language (Kartik et al., 2007), and a sender who controls what is revealed can steer a rational receiver's choice (Kamenica and Gentzkow, 2011). Citation counts can be inflated outright, as when editors pressure authors to cite their journal (Wilhite and Fong, 2012), and reward-maximizing learning agents with no built-in intent to deceive have been observed to distort their signals once distortion is profitable (Alon et al., 2023).

Existing designs differ in what the credit depends on: the group's return, the act of exploring, attention, verified gain, a penalty on checked lies, or past accuracy. Each fails to make truthful reporting the best choice

Credit from the shared return. Every agent receives the group's return, the default in cooperative multi-agent learning, where a shared reward does not say which agent contributed to it (Foerster et al., 2018). An explorer bears the full cost of exploring, while its own contribution to a return shared by many agents is hard to discern, so exploring and reporting are under-supplied. It also cannot tell a true report from an untruthful one, since both change the shared return only through what others later do with them.

Credit for keeping exploring. Explorers are credited for each exploration, whatever it yields. This sustains exploration, which TVC does not try to replace, but it is blind to what is reported: a true report, an exaggerated one and silence all earn the same, so honesty has no advantage, and credit is spent on explorations whose findings never reach the group. Untruthful reports then harm the group: they inflate apparent effects, as in the antidepressant trials above (Turner et al., 2008), and they add to avoidable waste in research (Chalmers and Glasziou, 2009).

Credit for only being followed (Popularity credit). Explorers are credited for attention, not accuracy. It encourages agents to report more more as they get rewarded more. An inflated claim attracts followers as readily as a real one, so exaggeration is profitable and a learner finds it.

Credit for only being accurate. Contradicting to popularity credit, explorers are credited only for the verified gain of those who rely on a claim. This leaves a true report that nobody uses with nothing, which is the incentive behind the file drawer, and a community can stall in a state where no one reports because no one listens. Such credit also never checks an unused claim, so truth, exaggeration and silence all earn the same. Accuracy-based remedies from the literature do not close this gap: peer prediction makes honesty an equilibrium

rather than a dominant strategy (Miller et al., 2005), and proper scoring rules require the outcome to be observed (Gneiting and Raftery, 2007).

A cost for lying alone. Reports that an audit finds false are fined (Becker, 1968). This lowers the agents' lying rate. Nevertheless, it is not enough either: a lie that is followed still collects the followers' credit, so the fine has to outbid it, and a true report that nobody follows still earns nothing, so truth is no better than silence. Moreover, a lie can be profitable outside the credit channel, as when a hyped claim attracts followers whose attention benefits the claimant, so the penalty must outweigh a temptation the designer does not control. With smooth lying costs, Kartik (2009) shows that truthful reporting is never an equilibrium, and that inflation shrinks with cost intensity but persists.

Reputation without credit. Receivers stop following senders who were wrong before, and give them nothing. This learns only from claims that were checked, and a claim that nobody follows is never checked, so unused lies leave a sender's reputation untouched. It also gives an honest explorer no reason to report: a new sender starts without trust and may not be followed, so a true report can go unused and earn nothing.

These failures set our goal. We ask for a credit rule under which reporting the truth is each agent's best action whatever the others do with the report, including when a lie or a withheld result carries a private gain the designer neither funds nor observes. Because receivers in learning systems, like readers of a literature, keep changing how they respond, the guarantee cannot rest on an equilibrium or on a model of receiver behavior; it must hold for every receiver rule, so truth must beat the response most favorable to a lie.

Our key idea is to reward only verified value and to make every claim sometimes face an audit. We propose Truthful Verified Credit (TVC), which can wrap a credit scheme with three parts that together reward truth at least as much as a lie in every case. Verified gain credit rewards the realized improvement of those who rely on a report, capped, and only when the report passes verification. A checked score bonus rewards every report that is checked, whether by a follower or by an independent random audit, so a true report, including a negative one, earns credit even when nobody acts on it. An audit fine is charged when an audit finds a report false, and we derive the fine needed to outweigh a lie's outside gain. TVC governs what is reported, not how much is explored, so it is meant to be combined with credit that encourages exploration.

We study TVC in two settings. In a multi-agent bandit, where agents search arms of unknown value on behalf of a group, popularity credit drives over 90% of reports to be exaggerated, while TVC brings it below one percent at comparable welfare. In a congestion setting built on a multi-agent pathfinding (MAPF) benchmark (Skrynnik et al., 2025), robots can post fake jams to clear their own routes: self-interested learners lie at 99% of opportunities without credit, TVC holds the rate near 0.2%, and throughput rises about threefold.

2 Related Work

Credit assignment problems in cooperative multi-agent systems. Credit assignment in cooperative multi-agent systems (MAS) asks how to credit each agent for its share of a shared return. Payoff design gives each agent a payoff that serves the group objective

(Wolpert and Tumer, 2001), counterfactual baselines estimate an agent's marginal contribution (Foerster et al., 2018), and value decomposition splits a team value into per-agent values (Sunehag et al., 2018; Rashid et al., 2018). These methods assume cooperative agents who report honestly. With mixed motives, as in sequential social dilemmas (Leibo et al., 2017), shaping the reward changes behavior: inequity aversion promotes cooperation (Hughes et al., 2018), social-influence rewards credit an agent for changing others' actions (Jaques et al., 2019), and gifting or learned incentives let agents reward one another (Lupu and Precup, 2020; Yang et al., 2020). These incentives reward being relied on, not being right, so an agent can raise its credit by exaggerating. This is an instance of reward hacking, where optimizing a proxy reward degrades the true objective (Amodei et al., 2016; Skalse et al., 2022). TVC is a rule for what to reward, and it can replace that credit.

Strategic communication. Conflicts of interest limit what cheap talk can convey (Crawford and Sobel, 1982). Receivers who take messages at face value are deceived by inflated language (Kartik et al., 2007), a sender who controls what is revealed can steer a rational receiver (Kamenica and Gentzkow, 2011), and with a smooth cost of lying, inflation falls but persists (Kartik, 2009). Learning agents with no intent to deceive distort their signals once distortion is profitable (Alon et al., 2023). These works analyze equilibria against a model of the receiver. We ask for dominance against every receiver rule.

Strategic experimentation and social learning. When agents share the results of costly experiments, they free-ride on one another (Bolton and Harris, 1999; Keller et al., 2005). A designer who controls what is recommended can induce myopic agents to explore while keeping each recommendation incentive compatible (Kremer et al., 2014; Mansour et al., 2015). Receivers who follow claims can turn an early wrong claim into a cascade (Banerjee, 1992; Bikhchandani et al., 1992), and reputation systems let them learn whom to trust (Resnick et al., 2000). In these models, what an explorer found is observed and not reported, so it cannot be misreported. In our game the explorers are the senders, and what they report can be false.

Truthful elicitation and verification. Proper scoring rules need the outcome to be observed (Gneiting and Raftery, 2007), and peer prediction (Miller et al., 2005; Shnayder et al., 2016; Feng et al., 2022; Gao et al., 2019) makes honesty an equilibrium and not a dominant strategy. With costly verification, the optimal mechanism checks only some reports, and a cost of lying lowers how often (Ben-Porath et al., 2014), in line with deterrence by probabilistic enforcement (Becker, 1968).

Verification, audits and penalties. Costly state verification starts from the idea that checking a report is costly (Townsend, 1979), and random auditing can be useful (Border and Sobel, 1987; Estornell et al., 2021). In algorithmic mechanism design, probabilistic verification lets a lie be caught with some probability, and the implementable truthful rules can be characterized (Caragiannis et al., 2012). These works study a principal who allocates after agents' reports and agents who respond optimally. Our receivers act by their own rule, and our senders learn.

Incentives for truthful reports. Mechanism design obtains dominant-strategy truthfulness when a planner picks the outcome that maximizes reported welfare and charges

each agent the externality it imposes (Vickrey, 1961; Clarke, 1971; Groves, 1973). That argument needs the planner to implement the outcome, but here receivers act by their own rule, so TVC keeps only the idea of crediting a claim by its effect on others. Peer prediction removes the need for ground truth by scoring a report against peers' reports (Prelec, 2004; Mandal et al., 2020), and incentive schemes for participatory sensing reward reports without observing the ground truth (Radanovic and Faltings, 2015). When agents learn, no-regret learning alone does not guarantee that they converge to truthful reporting, which holds only for a family of learners whose updates depend on cumulative rewards (Feng et al., 2022). More generally, designing against participants who run learning algorithms differs from designing against best responders (Braverman et al., 2018; Deng et al., 2019). Combining peer prediction with limited access to the ground truth (Gao et al., 2019) is close in spirit to our audit. A TVC audit checks a claim directly and does not use peers' reports, so it does not depend on how receivers behave.

3 Problem Setting

We consider a multi-agent bandit game with private payoff, extending from (Renault et al., 2025). There are N agents and K arms. Arm k has a value $v_k > 0$, drawn i.i.d. from a distribution F with finite mean and fixed within an episode. The mean value is $\bar{v} = \frac{1}{K}\sum_k v_k$, and all credit amounts below are in units of $\bar{v}$. Play lasts T rounds. At round t, agent i holds a set of known arms $\mathcal{K}_i^t \subseteq \{1, \dots, K\}$ with current best value $b_i^t = \max_{k \in \mathcal{K}_i^t} v_k$.

In each round, a free agent i chooses through its policy π_i to *explore*, pulling an unknown arm and adding it to $\mathcal{K}_i$, or to *exploit*, pulling its best known arm. Exploring costs $c \geq 0$. Agent i observes the payoff of its own pull only. On discovering arm k, agent i observes $y_i^t = v_k e^{\epsilon_i^t}$ with $\epsilon_i^t \sim \mathcal{N}(0, \sigma^2)$, drawn i.i.d. as observation noise, a random error that the explorer picks up when it discovers an arm. We write $\theta_i^t = v_k$ for the true value behind this observation. Pulling an arm yields its true value; only the discovery observation is noisy.

After discovering an arm, agent i may post a public claim $m_i^t \in \{\varnothing\} \cup \mathbb{R}_{>0}$, where $\varnothing$ is silence. Posting a claim costs $\xi \geq 0$. The claim is *truthful* if $m_i^t = y_i^t$. We call $\kappa_i^t = m_i^t / y_i^t$ the inflation rate. Each free agent j that did not explore then chooses whom to follow, $d_j^t = \rho_j(h_j^t, m^t) \in \{0\} \cup \{1, \dots, N\} \setminus \{j\}$, where 0 means nobody, h_j^t is j's private history and $m^t = (m_1^t, \dots, m_N^t)$. Following i commits j to i's arm for L rounds. The *receiver rule* ρ_j is an arbitrary function: it may be a fixed heuristic, a Bayesian update, a learning policy or an adversary. We assume no rationality and no equilibrium among receivers. The follower's gain over its own best is

$$g_{ji}^t = L\left(\theta_i^t - b_j^t\right)^+. \tag{1}$$

A receiver cannot check a claim before it acts on it; it learns θ_i^t exactly only by following.

Reward. A sender's payoff u_i^t is the true value of pulled arm, minus c when exploring.

A sender maximises a mixed-motive reward in which it weighs the others' mean payoff $\bar{u}_{-i}^t$ by the *prosocial weight* $\lambda \in [0, 1]$, where $\lambda = 0$ is a selfish sender and $\lambda = 1$ a fully cooperative one,

$$r_i^t = (1 - \lambda)\, u_i^t + \lambda\, \bar{u}_{-i}^t - \xi\, \mathbb{1}[m_i^t \neq \varnothing] + C_i^t, \tag{2}$$

Outside gain. A claim can also change the sender's reward outside the credit, because followers act on it. Relative to a truthful claim, a lie raises the liar's own payoff u_i^t by $G_1 \geq 0$,

e.g., the citations a researcher gains, and lowers the others' mean payoff $\bar{u}^t_{-i}$ by $D_1 \geq 0$, e.g., the time agents waste on a misleading approach, both per lie and in units of $\bar{v}$. Formally, $G_1 = \mathbb{E}[u^t_i \mid \text{lie}] - \mathbb{E}[u^t_i \mid \text{truthful claim}]$ and $D_1 = \mathbb{E}[\bar{u}^t_{-i} \mid \text{truthful claim}] - \mathbb{E}[\bar{u}^t_{-i} \mid \text{lie}]$, where the expectation is over how the receivers respond. A sender's payoff changes with a claim only through what others do, so $G_1 > 0$ needs a setting in which others' actions affect it. We take the largest values over receiver rules. The sender feels a share $1 - \lambda$ of its gain and a share λ of the damage, so a lie changes the part of its reward that does not come from the credit by the *outside gain*

$$\Gamma = (1 - \lambda)G_1 - \lambda D_1. \tag{3}$$

The designer neither observes nor provides Γ, and we assume $\Gamma \geq 0$. The posting cost ξ is borne by truth and lie alike, so it does not enter Γ; it matters only when a claim is compared with silence.

4 Truthful Verified Credit

With the problem defined in Section 3, we introduce TVC. Its three parts, verified gain credit, a checked score bonus and an audited fine, each answer one failure of reliance credit.

4.1 Verified Externality Credit

Popularity credit rewards a fake claim and a real one alike, because both attract followers. Let $\mathcal{F}^t_i = \{j : d^t_j = i\}$ be the *follow set* of sender i: the receivers j that follow i's claim in round t and so observe θ^t_i exactly. TVC instead credits the followers' realized gain Equation (1), and only for a claim that is verified:

$$\sum_{j \in \mathcal{F}^t_i} V^t_i \, \min\big(g^t_{ji},\, \bar{B}\bar{v}\big).$$

Here $V^t_i = \mathbb{1}\big[\, |m^t_i/\theta^t_i - 1| \leq \tau \,\big]$ is the verification gate: it equals 1 when the claim lies within a tolerance $\tau \geq 0$ of the true value θ^t_i, and 0 otherwise. The tolerance τ is set above the honest noise level ϵ, so a truthful claim passes the gate and any claim outside the tolerance is blocked. Every $j \in \mathcal{F}^t_i$ learns θ^t_i exactly on committing, so a followed claim ($\mathcal{F}^t_i \neq \emptyset$) is checked at no cost to the designer.

The gain g^t_{ji} is the externality of i's report on j relative to j staying put, as in a VCG mechanism (Vickrey, 1961; Clarke, 1971; Groves, 1973). A lie that is followed fails the gate V^t_i and earns no gain credit, so attracting followers with an inflated claim no longer brings a gain. The cap $\bar{B}\bar{v}$, with $\bar{B} > 0$ a design parameter in units of the mean value $\bar{v}$, limits what one follower can add, so a single lucky find cannot dominate the sender's return. Each receiver follows at most one claim, so the cap also bounds the cost: the gain term adds at most $N\bar{B}\bar{v}$ per round.

4.2 Checked Score Bonus

Gain credit alone leaves silence as a best response: a true claim that nobody follows earns nothing, exactly like staying silent. TVC therefore gives every checked claim a bonus that peaks at the truth,

$$\chi^t_i \, \eta \, \bar{v} \, S\big(m^t_i, \theta^t_i\big),$$

where $\eta > 0$ is the bonus weight and the score is a Gaussian in the log error with width $w > 0$,

$$S(m, \theta) = \exp\Big(- \frac{(\ln m - \ln \theta)^2}{2w^2} \Big).$$

The score is a bounded reward for a claim once the truth is revealed, and it behaves like a proper scoring rule (Gneiting and Raftery, 2007). We use the exponential of the squared log error as it is bounded in $[0, 1]$. The bonus costs at most $\eta\bar{v}$ per checked claim, and a lie's bonus is at most $\eta\bar{s}\bar{v}$. The log error matches the multiplicative noise and treats overstating and understating alike. The weight η sets the budget and must satisfy $p\eta > \xi$ (Appendix A), and the width w sets how fast the score falls and should be at least the observation noise σ, so that an honest claim is not scored down. A claim is *checked* when it is audited or followed,

$$\chi_i^t = \mathbb{1}[m_i^t \neq \varnothing] \, \max\big(a_i^t, f_i^t\big), \qquad f_i^t = \mathbb{1}\big[\mathcal{F}_i^t \neq \emptyset\big],$$

where f_i^t indicates that someone followed i, and a_i^t indicates a random audit of the claim, drawn independently of the claim and of the receivers,

$$a_i^t \sim \text{Bernoulli}(p), \qquad p \in (0, 1].$$

A claim is thus checked either by a follower or by a random audit, and the audit is what reaches claims nobody follows. Because the audit is independent of the claim and of the receivers, a true claim earns at least $p\,\eta\bar{v}$ in expectation however receivers behave, and this is strictly more than silence, which earns 0. This holds for a bad find as well, so reporting a low-value find is no longer a loss. The score is 1 at the truth and below $\bar{s} = \sup_{|m/\theta - 1| > \tau} S(m, \theta) < 1$ for every claim that fails the tolerance, so inflation cannot raise the bonus. For $\tau = w = 0.05$ the supremum over all claims is $\bar{s} \approx 0.62$, and it is far smaller when the lies available are far from the truth: 0.16 for a $\times 1.1$ lie and about 0 for a factor-two lie.

4.3 Audited Lying Cost

The first two parts leave a lie with a small payoff, the bonus up to $\eta\bar{s}\bar{v}$, and say nothing about an outside gain Γ, the net private benefit a lie brings its sender outside the credit channel, for example a false report that steers others away from something the sender wants for itself. TVC therefore fines every audited claim that fails verification, with a fine $\varphi \geq 0$ in units of $\bar{v}$:

$$a_i^t \, (1 - V_i^t) \, \mathbb{1}[m_i^t \neq \varnothing] \, \varphi \, \bar{v}.$$

The fine is charged only on audits. An audit has probability p, independent of the claim and the receivers, so a lie costs $p\varphi\bar{v}$ in expectation whatever the receivers do, and the fine can be sized against Γ without a model of the receivers (see Equation (5)).

Together, the credit to sender i in round t is

$$\begin{aligned} C_i^t = & \underbrace{\textstyle\sum_{j \in \mathcal{F}_i^t} V_i^t \min\big(g_{ji}^t, \bar{B}\bar{v}\big)}_{\text{verified gain credit}} \\ & + \underbrace{\chi_i^t \, \eta \, \bar{v} \, S\big(m_i^t, \theta_i^t\big)}_{\text{checked bonus}} \\ & - \underbrace{a_i^t \big(1 - V_i^t\big) \mathbb{1}\big[m_i^t \neq \varnothing\big] \, \varphi \, \bar{v}}_{\text{audited fine}} . \end{aligned} \tag{4}$$

The algorithm for learning with TVC per episode is shown in Algorithm 1, and the credit of one claim in Algorithm 2.

4.4 When TVC Is Useful

TVC is aimed at settings where four conditions hold together.

1. *Credit follows reliance.* Agents are credited for being followed or used, which rewards exaggeration and leaves a true report that nobody uses uncredited.
2. *Lying is profitable outside the credit.* Agents are not fully prosocial ($\lambda < 1$), so a lie can bring a private gain that no reliance credit removes by itself.
3. *Lying harms others.* A false report misleads those who act on it and wastes their effort, and the group loses once several agents lie. Where receivers can decode exaggeration at little cost, hype costs little, and TVC buys honest reports but not extra welfare.
4. *Claims can be audited at a cost.* Receivers cannot verify a claim before acting, but the designer can check a random sample at rate p.

One of our testbeds meets all four (Section 5). Other candidates, which we do not evaluate, are crowd-sourced traffic or robot-fleet reports, teams of language-model agents that build on one another's findings, and scientific credit through citations with sampled replication as the audit. When a condition fails, simpler designs may suffice: a shared return can work for nearly altruistic agents, and without any audit, peer prediction is the relevant regime.

5 Experiments

We study TVC in a multi-agent bandit, where agents search arms of unknown value on behalf of a group, and in a congestion setting built on a multi-agent pathfinding (MAPF) map (Skrynnik et al., 2025), where robots can post fake jams to clear their own routes.

5.1 Experiment 1: Multi-agent Bandit

Setup. $N = 10$ agents search $K = 50$ arms whose values are drawn i.i.d. from a lognormal distribution, in episodes of $T = 100$ rounds, and each agent starts knowing one random arm. In each round a free agent either explores a random arm, at a cost of $0.5\bar{v}$, or exploits its best known arm. An explorer may post a claim about its find: silence, or the true value scaled by $\times 1$ (truthful), $\times 1.1$, $\times 1.5$ or $\times 2$. Free agents that did not explore decode every

Algorithm 1 Learning with TVC (per episode)

Each sender decides to post a claim m_i^t or stay silent; draw an audit $a_i^t \sim \text{Bernoulli}(p)$ per claim
Receivers act on the claims by their own rule; record who relied on which claim
Mark a claim checked if it was audited or its outcome was observed by someone who relied on it
for each claim **do**
  Compute C_i^t by Algorithm 2
  Add C_i^t to the return of the decision that posted the claim
end for
Update each sender's policy with its learning rule

Algorithm 2 TVC credit of one sender in one round (the credit step of Algorithm 1)

Require: claim m_i^t or silence $\varnothing$; follow set $\mathcal{F}_i^t$ and gains g_{ji}^t; audit draw a_i^t; true value θ_i^t, revealed by a check; parameters $\tau, \bar{B}, \eta, w, \varphi$
Ensure: credit C_i^t
  if $m_i^t = \varnothing$ **then**
    return 0
  end if
  $\chi_i^t \leftarrow \max\big(a_i^t,\ \mathbb{1}[\mathcal{F}_i^t \neq \emptyset]\big)$ {checked by an audit or a follower}
  if $\chi_i^t = 0$ **then**
    return 0 {an unchecked claim earns nothing}
  end if
  $V_i^t \leftarrow \mathbb{1}\big[\,|m_i^t/\theta_i^t - 1| \le \tau\,\big]$ {verification gate}
  $C_i^t \leftarrow \eta\,\bar{v}\,S\big(m_i^t, \theta_i^t\big)$ {score bonus}
  if $V_i^t = 1$ **then**
    $C_i^t \leftarrow C_i^t + \sum_{j \in \mathcal{F}_i^t} \min\big(g_{ji}^t, \bar{B}\bar{v}\big)$ {verified gain credit}
  else if $a_i^t = 1$ **then**
    $C_i^t \leftarrow C_i^t - \varphi\,\bar{v}$ {fine for an audited false claim}
  end if
  return C_i^t

claim and follow the one with the highest decoded value, if it beats their own best, and stay committed for $L = 3$ rounds. We use two kinds of receiver, which differ only in how they decode. A *naive* receiver takes a claim at face value, so the decoded value of claim m_i is m_i. A *calibrated* receiver j keeps an estimate $\hat{\kappa}_{ji}$ of how much sender i inflates, starting at 1, and decodes the claim as $m_i/\hat{\kappa}_{ji}$. When j follows i it learns the true value of the find, and it updates $\hat{\kappa}_{ji} \leftarrow \hat{\kappa}_{ji} + 0.3\,(\kappa_i - \hat{\kappa}_{ji})$ with $\kappa_i = m_i/\theta_i$, so a sender that keeps inflating is decoded back toward its true value. Explore and report decisions are learned by tabular REINFORCE (learning rate 0.1, discount 0.9), with the agent's own best and the value of its find as states.

Credit designs. We compare no credit, popularity credit ($2\bar{v}$ per follower), accuracy credit (the follower's signed gain), TVC ($\tau = 0.05$, $\bar{B} = 8$, $\eta = 2$, $p = 0.2$, $\varphi = 8$, $w = 0.05$), a shared return (every agent receives the group's mean payoff), credit for keeping exploring (a flat $3\bar{v}$ per exploration, matched to TVC's spend), a lying cost alone (audited false claims fined φ, nothing else given) and popularity credit with that fine. Reputation alone is no credit with calibrated receivers.

Metrics. Inflation (share of reports that are exaggerated), report rate (reports per exploration), misled follows (follows that lead to a worse arm), welfare (own payoff per round, in units of $\bar{v}$) and exploration rate. We report the mean over the last 100 of 1000 episodes with 95% confidence intervals over 32 independent runs.

Results. Table 1 supports three findings. First, credit for being relied on encourages exaggeration, and calibrated receivers hide the cost without removing the cause. Popularity credit raises inflation above the level of no credit (94.0% against 75.1%), because the sender is credited for every follower whatever the claim is worth, so a large lie that attracts followers earns as much as a true report. With calibrated receivers the misled shrinks but inflation stays at 92.2%: the receivers learned to discount the lies, and the senders did not stop telling them. Accuracy credit helps only in part, because a claim that nobody follows earns nothing, true or false.

Second, honesty needs both a check on every claim and a reason to report. TVC removes inflation (0.3%) and keeps the agents reporting (report rate 99.6%), whereas a fine alone removes most inflation (3.4%) but leaves half of the explorers silent. A fine deters lies, and the checked bonus is what makes a truthful report worth posting. Adding the fine to popularity credit still leaves 38.4% inflation, because credit for being followed outweighs a fine that is charged only on audited claims.

Third, honest reports cost no welfare, but TVC does not create exploration. Its welfare equals that of popularity credit and exceeds no credit, so honesty is free here. The higher welfare of credit for keeping exploring (4.28) comes from exploring more (15.0% against 3.0%), and it leaves inflation at 74.7%.

TVC wraps a credit scheme. TVC can sit on top of a credit that rewards a claim, e.g., popularity credit, accuracy credit, a social-influence credit or keep-exploring credit. That credit is passed on only when the claim passes the verification gate. A claim therefore earns additional amount of $V_i\mathbb{H}_i$ with $\mathbb{H}_i$ is the host credit. A lie then keeps none of the host credit, and the one-round condition on p and φ does not depend on the host, if a truthful claim's host credit is bounded below. Credit for keeping exploring is wrapped too, but it ignores the report, so it needs no gate and has a single TVC arm. A shared return rewards the group's outcome and not a claim, so we do not wrap it. Table 2 compares each host alone, with TVC's terms added without the gate, and with it. The gate is what removes the host's incentive to lie: without it, a followed lie still collects popularity credit and 17.9% of reports stay inflated with naive receivers, against 0.2% with it. The two credits also do not interfere. With credit for keeping exploring, TVC leaves the exploration rate unchanged within the confidence intervals (15.0% against 14.3% with naive receivers) while inflation falls from about 75% to under 1%, so TVC fixes what is reported and the host keeps what it encourages.

Table 1: Multi-agent bandit: credit designs compared. Inflation is the percentage of reports that are exaggerated, report is the percentage of explorations that are reported, misled is the percentage of follows that lead to a worse arm, and explore is the percentage of agent-rounds spent exploring. Welfare is the mean own payoff per agent per round, in units of the mean arm value. With calibrated receivers, *None* is reputation only. *Keeping exploring* gives a flat $3\bar{v}$ per exploration, *Lying cost alone* fines audited false claims and gives nothing else, and *Shared return* gives every agent the group mean payoff.

Credit design	Inflation (%)	Report (%)	Misled (%)	Welfare ($\bar{v}$)	Explore (%)
Naive receivers					
None	75.1 ± 0.4	80.2 ± 0.3	25.5 ± 1.0	3.486 ± 0.104	2.7 ± 0.2
Popularity	94.0 ± 0.5	94.8 ± 0.4	43.5 ± 1.1	4.036 ± 0.054	4.5 ± 0.2
Accuracy	68.9 ± 2.4	89.8 ± 0.4	13.7 ± 1.2	3.867 ± 0.095	2.9 ± 0.2
Shared return	71.9 ± 5.4	80.1 ± 4.7	32.7 ± 4.8	4.055 ± 0.145	$\mathbf{23.3 \pm 3.0}$
Keeping exploring	74.7 ± 1.1	79.1 ± 1.2	41.3 ± 1.8	$\mathbf{4.280 \pm 0.083}$	15.0 ± 2.5
Lying cost alone	3.4 ± 0.7	50.0 ± 2.9	1.4 ± 0.3	3.177 ± 0.114	2.4 ± 0.2
Popularity + fine	38.4 ± 1.6	85.0 ± 2.4	34.4 ± 1.6	3.859 ± 0.070	3.4 ± 0.2
TVC	$\mathbf{0.3 \pm 0.1}$	$\mathbf{99.6 \pm 0.1}$	$\mathbf{0.2 \pm 0.1}$	4.028 ± 0.079	3.0 ± 0.2
Calibrated receivers					
None	75.1 ± 0.5	80.1 ± 0.4	7.6 ± 0.5	3.528 ± 0.126	2.5 ± 0.3
Popularity	92.2 ± 0.5	93.4 ± 0.5	0.9 ± 0.4	4.023 ± 0.078	3.2 ± 0.2
Accuracy	85.1 ± 1.3	90.6 ± 0.4	3.3 ± 0.5	3.993 ± 0.097	3.1 ± 0.3
Shared return	68.0 ± 6.8	83.6 ± 3.4	3.3 ± 0.7	4.118 ± 0.159	$\mathbf{23.7 \pm 3.6}$
Keeping exploring	74.8 ± 1.2	79.2 ± 1.1	9.1 ± 0.6	$\mathbf{4.358 \pm 0.079}$	15.1 ± 2.6
Lying cost alone	3.5 ± 0.7	50.4 ± 3.6	1.3 ± 0.2	3.045 ± 0.128	2.2 ± 0.3
Popularity + fine	33.4 ± 2.3	85.4 ± 2.5	2.5 ± 0.4	3.877 ± 0.094	2.5 ± 0.2
TVC	$\mathbf{0.4 \pm 0.1}$	$\mathbf{99.6 \pm 0.2}$	$\mathbf{0.2 \pm 0.1}$	3.963 ± 0.081	2.9 ± 0.2

5.2 Experiment 2: MAPF with Fake Jams

Setup. We use POGEMA's lifelong pathfinding task (Skrynnik et al., 2025) with soft collisions on a 3×3 layout of 5×5 rooms joined by 1-cell doorways, with 12 robots, observation radius 3 and 256 steps. Robots move by greedy descent on a distance map, so the only learned decision is whether to post a claim that a doorway they can see is jammed. We define the ground truth that a doorway is jammed when at least 3 robots are within 2 walking steps of it. Receivers that believe a claim avoid that doorway for 20 steps. A claim is checked by a robot that passes the doorway, or by an audit. Naive receivers take claims at face value, and sceptical receivers learn per-sender trust from the checks they see (learning rate 0.3).

Mixed motive. A robot's reward is $(1 - \lambda)$ times its own deliveries plus λ times the others' mean deliveries, minus a cost per claim posted, plus credit. A lie, a fake jam on its own route, gains the liar when believers leave that route and costs the others their detours. The per-believer gain, harm and benefit that enter the learning signal are calibrated from short runs with hand-set policies, and the temptation is therefore a modelling choice, not an emergent property.

Table 2: TVC on top of a credit scheme in the multi-agent bandit, with naive and calibrated receivers. Inflation is the percentage of reports that are exaggerated, welfare is the mean own payoff per agent per round in units of $\bar{v}$, and explore is the percentage of agent-rounds spent exploring. Influence is a simplified social-influence credit in the spirit of Jaques et al. (2019). Gated means the host credit is passed on only for claims that pass the verification gate, and ungated means TVC's bonus and fine are added to the host credit as they are.

	Naive receivers			Calibrated receivers		
Credit design	Inflation (%)	Welfare	Explore (%)	Inflation (%)	Welfare	Explore (%)
None	75.1 ± 0.4	3.49 ± 0.10	2.7 ± 0.2	75.1 ± 0.5	3.53 ± 0.13	2.5 ± 0.3
Popularity	94.0 ± 0.5	4.04 ± 0.05	4.5 ± 0.2	92.2 ± 0.5	4.02 ± 0.08	3.2 ± 0.2
+ TVC, ungated	17.9 ± 1.3	4.03 ± 0.09	4.2 ± 0.6	11.2 ± 1.2	3.99 ± 0.08	3.0 ± 0.2
+ TVC, gated	0.2 ± 0.0	4.15 ± 0.07	3.4 ± 0.2	0.3 ± 0.0	4.06 ± 0.08	3.1 ± 0.2
Accuracy	68.9 ± 2.4	3.87 ± 0.10	2.9 ± 0.2	85.1 ± 1.3	3.99 ± 0.10	3.1 ± 0.3
+ TVC, ungated	3.2 ± 0.9	4.01 ± 0.09	3.2 ± 0.2	6.3 ± 1.2	4.05 ± 0.10	3.2 ± 0.3
+ TVC, gated	0.4 ± 0.1	4.05 ± 0.10	3.2 ± 0.2	0.3 ± 0.1	4.12 ± 0.07	3.2 ± 0.2
Influence	91.8 ± 0.8	4.03 ± 0.07	4.6 ± 0.1	92.3 ± 0.7	4.07 ± 0.08	3.4 ± 0.2
+ TVC, ungated	5.0 ± 1.1	4.18 ± 0.06	4.0 ± 0.2	4.0 ± 1.0	4.18 ± 0.08	3.6 ± 0.2
+ TVC, gated	0.2 ± 0.1	4.20 ± 0.06	3.7 ± 0.2	0.3 ± 0.1	4.20 ± 0.08	3.6 ± 0.2
Keeping exploring	74.7 ± 1.1	4.28 ± 0.08	15.0 ± 2.5	74.8 ± 1.2	4.36 ± 0.08	15.1 ± 2.6
+ TVC	0.1 ± 0.0	4.63 ± 0.06	14.3 ± 2.2	0.2 ± 0.1	4.63 ± 0.06	14.7 ± 2.0
TVC alone	0.3 ± 0.1	4.03 ± 0.08	3.0 ± 0.2	0.4 ± 0.1	3.96 ± 0.08	2.9 ± 0.2

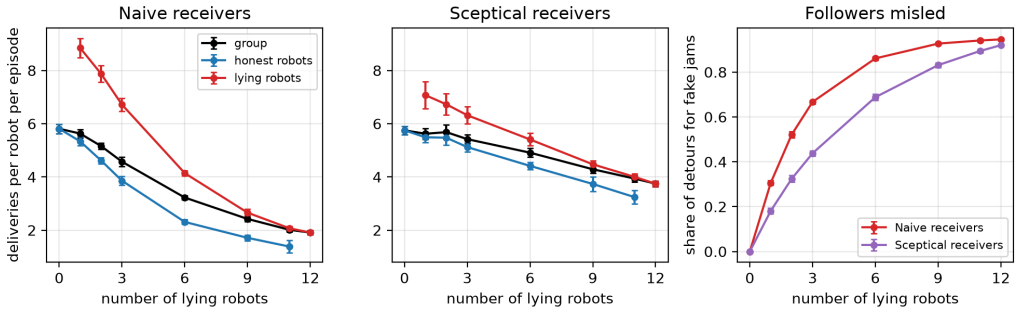


Figure 1: Lying harms the group in the congestion setting. The first k of 12 robots post fake jams and the rest report real jams, with fixed policies and no credit. Left and middle: deliveries per robot per episode for the group, the honest robots and the lying robots, with naive and with sceptical receivers. Right: share of detours that were for fake jams.

Credit designs. No credit, popularity credit (1 per believer), accuracy credit (the steps a believer saves, negative for a fake jam), a lying cost alone (audited false claims are fined and nothing else is given), popularity credit plus the same fine, TVC ($\tau = 0.05$, $\bar{B} = 8$, $\eta = 2$, $p = 0.2$, $\varphi = 15$), reputation (no credit, sceptical receivers) and a shared return ($\lambda = (N-1)/N$).

Table 3: POGEMA lifelong MAPF with every credit design and tabular REINFORCE claim learners ($\lambda = 0$, 10 seeds, mean ± 95% CI). Reputation is no credit with sceptical receivers.

Design	Lie rate (%)	Useful (%)	Throughput
None	99.2 ± 0.0	44.7 ± 1.5	0.080 ± 0.002
Popularity	99.9 ± 0.0	94.6 ± 0.8	0.037 ± 0.001
Accuracy	0.1 ± 0.0	1.2 ± 0.6	0.061 ± 0.005
Lying cost alone	0.2 ± 0.0	34.8 ± 0.9	0.240 ± 0.008
Popularity + fine	47.0 ± 7.5	94.4 ± 1.2	0.059 ± 0.002
Reputation	80.1 ± 1.3	41.0 ± 1.5	0.124 ± 0.008
Shared return	6.3 ± 0.1	92.2 ± 1.2	0.249 ± 0.006
TVC	$\mathbf{0.2 \pm 0.1}$	$\mathbf{98.8 \pm 0.7}$	$\mathbf{0.266 \pm 0.007}$

Metrics. We report the lie rate (the share of opportunities on which a robot posts a fake jam on its own route), the useful-report rate (the share of real jams elsewhere that it reports), POGEMA's throughput (deliveries per step, summed over the 12 robots) as the mean over the last 50 of 150 episodes, with 95% confidence intervals over 10 seeds.

Lying harms the group. Hype is a social dilemma, which is why it is worth deterring: a lie helps the liar and costs the group. We check this with fixed claim policies instead of learned ones, the first k of the 12 robots posting a fake jam whenever a doorway on their own route is free, the others reporting real jams, and no credit. A single liar gains 3.5 deliveries over its honest peers while the others lose 0.5, so one lie mostly moves deliveries to the liar, but the cost to the group grows with the number of liars: deliveries per robot fall from 5.8 with no liars to 1.9 with all 12 (Figure 1, naive receivers). Sceptical receivers weaken the effect (3.7 with 12 liars, and a liar's advantage of 1.6) without removing it, so detecting lies after the fact is not enough here.

TVC removes lying and raises throughput. Table 3 shows that TVC is the only design that removes lying while keeping the robots reporting, and that throughput rises as a result. Without credit, selfish robots lie at almost every opportunity (99.2%) and report fewer than half of the real jams (44.7%), so what reaches the group is mostly false. Popularity credit makes the robots report real jams (94.6%) but not stop lying (99.9%), and throughput falls (0.080 to 0.037): credit for being followed rewards fake and real jams alike, so more reports mean more misleading ones. TVC removes lying (0.2%), keeps useful reports high (98.8%) and raises throughput to 0.266, 3.3 times that of no credit.

The other designs show that each part of TVC has its own job. A fine alone stops lying (0.2%) but the robots report only 34.8% of the real jams, because it punishes lies without rewarding truth; the checked bonus restores reporting and adds 11% throughput. The same fine on top of popularity credit still leaves 47.0% lying, because credit for being followed outweighs it. Reputation alone cannot deter lies that are never checked (80.1% lying), and accuracy credit silences the robots (1.2% useful reports), since a fake jam earns negative credit and a true jam that nobody uses earns nothing. A shared return comes closest (6.3% lying, throughput 0.249), because it makes each robot bear the group's loss, but it needs every robot to weigh the others' deliveries as much as its own.

6 Conclusion

Conclusion. Credit that rewards agents for being relied on rewards attention, not accuracy, and learning agents respond by exaggerating. We proposed truthful verified credit (TVC), which gives credit only for verified value, adds a bonus for every checked claim, and fines audited false claims. In both experiments TVC nearly eliminates exaggeration and keeps agents reporting. Its welfare matches popularity credit when receivers can decode exaggeration cheaply, and it is well above no credit and popularity credit when receivers act on claims. Because TVC gates a host credit by the same verification test, it can sit on top of popularity, accuracy or exploration credit and keep reports honest while the host keeps encouraging what it encourages, such as exploration. TVC is called for when credit follows reliance, lying is profitable outside the credit, lying harms others, and a sample of claims can be audited. When agents weigh the others' payoffs almost as much as their own, a shared return alone comes close.

Limitations. TVC elicits truth, not usefulness. It does not raise exploration by itself. The fine rule is derived for a sender that maximises expected reward in a single round, and it assumes that checks reveal the truth exactly, that audits are independent, and that receivers never follow a claim that cannot help them. Truthful reporting over many rounds of learning is supported by our experiments and not by a guarantee. Our testbeds are simulated, and the outside gain of a lie is calibrated from short runs, not an emergent property. Future work includes noisy verification and fleets with real data.

Appendix A. Analysis of Truthful Reporting in MAB with Private Payoff

Proposition 1 (Truthful reporting is interim strictly dominant in one round) *In one round, truthful reporting is an interim strictly dominant strategy for the sender with reward defined in Equation* (2) *if* $p(\eta + \varphi) > \eta\,\bar{s} + \Gamma$ *and* $p\eta > \xi$.

Proof Let us reconsider the problem introduced in Section 3. We relax the problem into a single round with one sender. For brevity, we will drop the "t" notation. In this case, a sender i found an arm, and choose to make a report $m \in \{\varnothing\} \cup \mathbb{R}_{>0}$. A report m is truthful if $m = \theta$ and untruthful otherwise, whether exaggerated ($m > \theta$) or understated ($m < \theta$), and $m = \varnothing$ is silence. Everyone else is collapsed into an arbitrary *opponent profile*, denoted σ_{-i}: any map from the posted claim to a distribution over their actions.

We assume the follows.

A1 *Separability.* Given the action profile, u_i does not depend on m_i. Claims affect i's reward only through C_i and through receivers' future behaviour.

A2 *Unforgeable verification.* On checked claims the designer learns θ_i exactly, and followers cannot misreport it.

A3 *Independent audits.* a_i is independent of (m, d, h).

A4 *Untruthful reports are detected.* Every untruthful report, whether exaggerated or understated, misses the tolerance: $m \neq \theta \Rightarrow |m/\theta - 1| > \tau$.

Table 4: Reward r of a sender i under TVC in one round, based on Equation (2), in units of $\bar{v}$. Each row is what happens to the claim: whether some receiver follows it and whether the random audit checks it. Each column is what the sender does: report truthfully ($m = \theta$), report untruthfully ($m \neq \theta$, exaggerated or understated) or stay silent ($m = \varnothing$). Each cell is Equation (2) evaluated with the credit of Equation (4) for the case in its row and column, with $S = S(m, \theta)$. Every claim bears the posting cost ξ. An untruthful report never passes the verification gate, so it earns no gain credit, and it gains the outside gain Γ only when it is followed, because only then do believers act on it. We take the outside effect of a truthful claim as zero, since a positive effect only favours truth. Silence posts no claim, so it bears no cost and earns 0.

Case	Truth	Untruthful	Silence
Followed, audited	$\min(g, \bar{B}) + \eta - \xi$	$\Gamma + \eta S - \varphi - \xi$	0
Followed, not audited	$\min(g, \bar{B}) + \eta - \xi$	$\Gamma + \eta S - \xi$	0
Not followed, audited	$\eta - \xi$	$\eta S - \varphi - \xi$	0
Not followed, not audited	$-\xi$	$-\xi$	0

A5 *Truthful reports are individually rational.* Setting credit aside, reporting the truth is weakly preferred to silence $O(\theta) \geq O(\varnothing)$. This holds when receivers never follow a truthful claim at a loss to themselves and following does not lower the sender's own payoff.

Table 4 shows the reward r_i of a sender under TVC in one round.

Fix any opponent profile σ_{-i}, and write $O(m)$ for the expected value of $(1 - \lambda)u_i + \lambda\bar{u}_{-i}$ when the sender reports m. By separability (A1), $O(m)$ depends on m only through how the opponents respond, so $O(m) \leq O(\theta) + \Gamma$ for every untruthful report m, and $O(\theta) \geq O(\varnothing)$ by A5. The audit is independent of the claim and the receivers (A3), so a non-silent claim is checked with probability at least p, and a report that fails the gate is fined with probability exactly p. By *interim strict dominance* we mean that, for every true value θ and every opponent profile σ_{-i}, the expected reward of truthful reporting exceeds that of every other report. The sender knows its own find θ when it reports, but not yet the audit draw or the receivers' response, and the expectation is over those two. It does not require truth to be best in every realisation, and it is not (silence is better in the last row of Table 4). Write $\bar{r}(m)$ for the expected reward of report m.

A truthful claim passes the gate and its gain credit is nonnegative. It earns the bonus η whenever it is checked, which happens with probability at least p. So, whatever the receivers do,

$$\bar{r}(\theta) \;\geq\; O(\theta) - \xi + p\eta.$$

By A4 an untruthful report fails the gate, so it earns no gain credit and is fined φ when audited. Its bonus is at most $\eta\bar{s}$, because its score is at most $\bar{s}$ and it is checked at most always. So, whatever the receivers do,

$$\bar{r}(m) \;\leq\; O(\theta) + \Gamma - \xi + \eta\bar{s} - p\varphi.$$

Silence bears no cost and cannot be followed or audited, so $\bar{r}(\varnothing) = O(\varnothing) \leq O(\theta)$.

Subtracting, for every value θ, every untruthful report $m \neq \theta$ and every σ_{-i},

$$\bar{r}(\theta) - \bar{r}(m) \;\geq\; p(\eta + \varphi) - \eta\bar{s} - \Gamma \;>\; 0 \qquad \text{and} \qquad \bar{r}(\theta) - \bar{r}(\varnothing) \;\geq\; p\eta - \xi \;>\; 0,$$

both by the conditions of Theorem 1. Neither bound depends on m or on σ_{-i}, so truthful reporting is interim strictly dominant. ■

Proposition 2 (Truthful reporting when later rounds matter) *Let the sender discount later rounds by $\gamma \in [0, 1)$, and let x be the state of the receivers: what they remember about the sender, their commitments and what they know. Write $r(x, \theta, m)$ for the expected reward of report m in the current round when the find is θ, $V^T(x)$ for the expected discounted reward of reporting truthfully from now on, and x' for the next state. Define*

$$Q^T(x, \theta, m) = r(x, \theta, m) + \gamma\, \mathbb{E}\big[V^T(x') \mid x, \theta, m\big],$$

the margin of one round

$$\mu = \min\big\{\, p(\eta + \varphi) - \eta\bar{s} - \Gamma,\;\; p\eta - \xi \,\big\},$$

and the future advantage of a lie

$$K = \gamma \sup_{x,\theta,\, m\neq\theta} \Big(\mathbb{E}\big[V^T(x') \mid x, \theta, m\big] - \mathbb{E}\big[V^T(x') \mid x, \theta, \theta\big] \Big),$$

where $m \neq \theta$ includes silence. If $\mu > K$, then reporting the truth is the unique optimal policy of the sender at every state x.

Proof Fix a state x and a find θ. By the argument of Proposition 1, applied in state x, the expected reward of the truth exceeds that of any other report by at least μ: by $p(\eta + \varphi) - \eta\bar{s} - \Gamma$ for an untruthful report, and by $p\eta - \xi$ for silence. By the definition of K, the future can lower this advantage by at most K. Hence

$$\begin{aligned} Q^T(x, \theta, \theta) - Q^T(x, \theta, m) &= \big[r(x, \theta, \theta) - r(x, \theta, m)\big] - \gamma\big[\, \mathbb{E}[V^T(x') \mid x, \theta, m] - \mathbb{E}[V^T(x') \mid x, \theta, \theta]\big] \\ &\geq \mu - K \\ &> 0 \qquad \text{for every } m \neq \theta. \end{aligned}$$

So the truth is the best report in every state when later values are those of the truthful policy, which means that V^T satisfies the Bellman optimality equation, $V^T(x) = \mathbb{E}_\theta \max_m Q^T(x, \theta, m)$. The Bellman operator is a contraction for $\gamma < 1$ and bounded rewards, so it has one fixed point, and V^T equals the optimal value. The inequality is strict, so no other report is optimal at any state. ■

The bound has to restrict K, since a receiver that punishes every truthful claim for many rounds makes K arbitrarily large. Γ must also cover the whole effect window of a claim, for example the 20 steps for which believers avoid a doorway.

Table 5: Lie rate when the fine keeps the expected fine fixed, $\varphi = 0.90/p$, twice break-even ($\lambda = 0$, 10 seeds, mean ± 95% CI, last 50 of 150 episodes).

Audit rate p	1	0.5	0.2	0.1	0.05	0.02	0.01
Fine φ	0.9	1.8	4.5	9.0	18	45	90
Lie rate (%)	0.9 ± 0.1	0.9 ± 0.1	0.9 ± 0.3	1.5 ± 1.0	45.1 ± 4.9	91.0 ± 5.9	97.9 ± 2.0

Appendix B. Sizing the Fine and Audit Rate

We size the fine against the outside gain Γ of a lie (Equation (3)) and the posting cost ξ (Equation (2)).

The rule. By the analysis above, truth earns at least $p\eta\bar{v}$ from the credit, and a lie earns at most $(\eta\bar{s} - p\varphi)\bar{v}$ from the credit plus Γ outside it, and both bear the posting cost ξ. Truth stays strictly better than a lie when

$$\varphi > \varphi_{\text{lo}} = \frac{\eta\,\bar{s} + \Gamma}{p} - \eta. \tag{5}$$

A larger outside gain needs a larger fine, and so do fewer audits, in proportion to $1/p$. A robot at a free doorway in the congestion game has nothing true to report, so its alternative to a lie is silence, which bears no posting cost, and ξ no longer cancels. The condition is then $p\varphi > \Gamma - \xi$ when $\bar{s} \approx 0$, and in the experiments φ_{lo} denotes $(\Gamma - \xi)/p$.

Fine and audit rate trade off. The product $p\varphi$ is the expected fine, and it has to exceed the net outside gain of a lie. In the congestion experiment this gain is $\Gamma - \xi = 0.450$ in units of $\bar{v}$ and $\bar{s} \approx 0$. So at audit rate p the break-even fine is $0.450/p$, and for a fine φ the break-even audit rate is $0.450/\varphi$: doubling the fine halves the audits needed.

Lowering the audit rate. We test this at $\lambda = 0$ (Figure 2). First we fix the fine at 9.0, four times the break-even fine at $p = 0.2$, the audit rate of the main experiments. Lying should then stop at $p^* = 0.450/9.0 = 0.05$ (dashed line). Lying is 99% at $p \le 0.02$, 62% at $p = 0.05$ and 1.5% at $p = 0.1$, and from $p = 0.1$ upward throughput stays at about 0.26 while audits per step fall from 0.17 at $p = 1$ to 0.018.

Then we raise the fine as the audit rate falls, setting it to twice break-even, $\varphi = 0.90/p$, so that the expected fine is the same at every audit rate (Table 5). Surprisingly, this does not help low audit rates: lying is 98% at $p = 0.01$ and 45% at $p = 0.05$, because a learner then sees too few audited claims, about two per episode across the fleet at $p = 0.01$. The expected fine is necessary but not sufficient, and in this testbed the audit rate has a practical floor near 0.1 that the fines we tried did not remove.

From this experiment, we found the safe choice for the fine is at break-even value φ_{lo} and rate value ≥ 0.1.

Appendix C. Notation

Table 6 lists the symbols used in the paper. Amounts of credit are in units of the mean arm value $\bar{v}$.

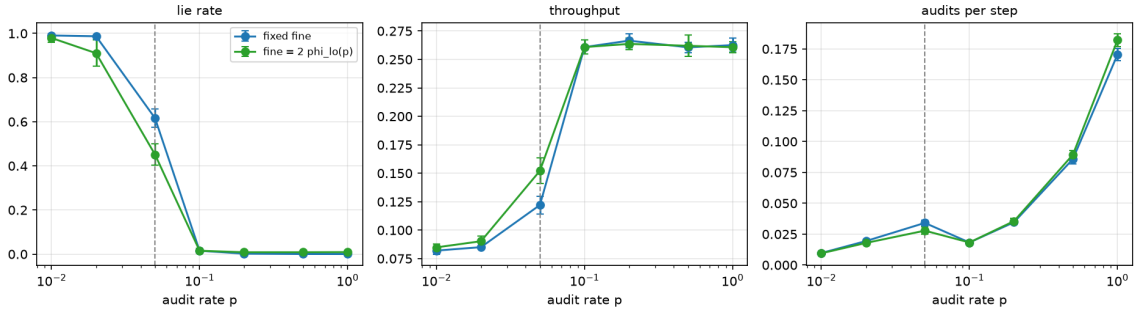


Figure 2: TVC against the audit rate p ($\lambda = 0$, 10 seeds, 95% CIs). Blue: the fine fixed at 9.0, with the predicted threshold $p^* = 0.05$ dashed. Green: the fine set to $0.90/p$. Audits per step count every audited claim, true or false, so they fall once the robots stop lying and post fewer claims.

Appendix D. Experimental Details

This appendix gives the hyperparameters, how the temptation of a lie is calibrated, and the seeds, statistics and compute behind Section 5.

D.1 Hyperparameters

Tables 7 and 8 list every setting of the two experiments. Credit amounts are in units of the mean arm value $\bar{v}$ in the bandit, and in deliveries in the congestion setting.

D.2 Calibrating the Temptation

The congestion setting needs the per-believer gain, harm and benefit of a claim, which enter the sender's learning signal (Equation (2)). We measure them before training, once for each layout, with no credit and with hand-set claim policies on common random seeds, 30 episodes each. In every policy only robot 0 is the sender and the other robots stay silent, except in the last. The *silent* policy never posts. The *reporter* posts true jams elsewhere, and the *liar* posts fake jams on its own route. In *all-liars* every robot posts fake jams on its own route. Writing $T_\bullet$ for the mean deliveries per robot under policy $\bullet$ and $b_\bullet$ for the number of believers of the sender's claims,

$$\text{own gain} = \frac{T^{(0)}_{\text{liar}} - T^{(0)}_{\text{silent}}}{b_{\text{liar}}}, \qquad \text{benefit} = \frac{\bar{T}^{(-0)}_{\text{reporter}} - \bar{T}^{(-0)}_{\text{silent}}}{b_{\text{reporter}}},$$

$$\text{harm} = \frac{N\,(\bar{T}_{\text{silent}} - \bar{T}_{\text{all-liars}}) + \text{own gain} \cdot b_{\text{all-liars}}}{(N-1)\,b_{\text{all-liars}}},$$

where (0) is robot 0 and (-0) the mean of the other robots. We measure the harm with all robots lying because a single liar among silent robots barely hurts: traffic spreads out. These per-believer values give Γ in Equation (3), and from it the fine threshold φ_{lo} in Equation (5) with $\bar{s} \approx 0$.

Table 6: Notation.

Symbol	Meaning
Environment	
N, K, T	number of agents, arms and rounds
v_k, F, $\bar{v}$	value of arm k, its distribution, and the mean value (unit of credit)
$\mathcal{K}_i^t$, b_i^t	arms known to agent i and the value of its best known arm
c	cost of exploring an unknown arm
u_i^t, π_i	own payoff of agent i and its policy
y_i^t, ϵ_i^t, σ	noisy observation $y_i^t = v_k e^{\epsilon_i^t}$, its noise, and the noise level
θ_i^t	true value of the arm that i reports on
Claims and reliance	
m_i^t, $\varnothing$	claim of sender i and silence
κ_i^t	inflation factor m_i^t/y_i^t (1 means truthful)
d_j^t, ρ_j, h_j^t	whom receiver j follows, its receiver rule, and its history
L	number of rounds a follower stays committed
g_{ji}^t	gain of follower j from i's claim, $L(\theta_i^t - b_j^t)^+$
$\mathcal{F}_i^t$, f_i^t	follow set $\{j : d_j^t = i\}$ and the indicator that it is non-empty
TVC	
τ	tolerance for verification
V_i^t	verification gate, $\mathbb{1}[\lvert m_i^t/\theta_i^t - 1\rvert \le \tau]$
$\bar{B}$	cap on gain credit per follower (in units of $\bar{v}$)
η, w	bonus weight and score width
$S(m,\theta)$, $\bar{s}$	score of a claim and the best score of any claim outside the tolerance
a_i^t, p	random audit and its probability
χ_i^t	claim is checked (audited or followed)
φ, φ_{lo}	fine for an audited false claim and the smallest fine that keeps truth dominant
C_i^t	TVC credit to sender i
Outside gain	
λ	prosocial weight on the others' mean payoff in the reward (0 selfish, 1 fully cooperative)
r_i^t, $\bar{u}_{-i}^t$	total reward of sender i and the mean payoff of the other agents
G_1, D_1	liar's gain and damage to the others from one lie
ξ	cost of posting a claim
Γ	outside gain of a lie over a truthful claim, $(1-\lambda)G_1 - \lambda D_1$
Δ	margin by which truth beats a lie, $p(\eta+\varphi) - \eta\bar{s}$

D.3 Seeds, Statistics and Compute

In the bandit, the 32 runs are independent policies and arm values simulated as one batch from one random generator (seed 0). The confidence interval is $1.96\, s/\sqrt{32}$, where s is the standard deviation of the per-run mean over the last 100 episodes. In the congestion setting, each seed trains from scratch, the metric of a seed is its mean over the last 50 of 150 episodes, and the confidence interval is the t-interval over seeds. All runs use the CPU only (an Apple M4 Max with 14 cores and 36 GB of memory), with seeds and variants distributed over a process pool.

Table 7: Multi-agent bandit settings.

Setting	Value
Agents N, arms K, rounds T	10, 50, 100
Episodes, independent runs	1000, 32
Exploration cost c	$0.5\bar{v}$
Commitment L	3 rounds
REINFORCE learning rate, discount	0.1, 0.9
Baseline step size	0.05
Calibrated receivers' learning rate	0.3
Popularity credit	$2\bar{v}$ per follower
Accuracy credit	$1\times$ follower's signed gain
Keeping-exploring credit	$3\bar{v}$ per exploration
TVC: τ, $\bar{B}$, η, w, p, φ	0.05, 8, 2, 0.05, 0.2, 8

Table 8: MAPF fake-jam settings. DQN and PPO learn independently for each robot, with one network (DQN) or an actor and a critic (PPO) per robot; rewards are clipped to $[-10, 10]$ for all three learners.

Setting	Value
Robots N, observation radius, steps	12, 3, 256
Episodes, seeds	150, 10
Jam radius, jam queue	2 steps, 3 robots
Avoidance duration, claim lifetime	20, 20 steps
Sceptical receivers' learning rate	0.3
Cost of posting a claim ξ	0.05
Prosocial weight λ	0 (main), swept in $\{0, \ldots, 1\}$
REINFORCE learning rate	0.1
Baseline step size, advantage clip	0.05, $[-10, 10]$
DQN: learning rate, replay buffer, batch, steps per episode	10^{-3}, 10,000, 64, 10
DQN: exploration rate ε	$1 \to 0$ over 100 episodes
PPO: learning rate, clip, epochs per episode, entropy bonus	$3 \cdot 10^{-4}$, 0.2, 10, 0.01
DQN, PPO: hidden layers per network	2×32 (ReLU, tanh)
Popularity credit	1 per believer
Accuracy credit	$1\times$ steps saved
TVC: τ, $\bar{B}$, η, p, φ	0.05, 8, 2, 0.2, 15

Appendix E. Additional Experiments

E.1 Other Claim Learners

Each robot learns its claim policy independently, with tabular REINFORCE (the default in Section 5), a DQN or PPO. A claim decision is one step long, so its return is its reward: the DQN regresses a Q-network onto it from a replay buffer, and PPO uses the reward minus a learned value as its advantage. DQN and PPO use untuned default hyperparameters (Appendix D). The REINFORCE results are in Table 3, and Table 9 gives the DQN and

Table 9: POGEMA lifelong MAPF with every credit design and DQN and PPO claim learners ($\lambda = 0$, 10 seeds, mean ± 95% CI, metrics as defined in Section 5.2). DQN's exploration rate falls to zero over the first 100 episodes, so its lie rate is that of its greedy policy. Reputation and shared return were run for REINFORCE only.

Design	Lie rate (%)	Useful (%)	Throughput
DQN			
None	100.0 ± 0.0	0.0 ± 0.0	0.090 ± 0.001
Popularity	100.0 ± 0.0	98.8 ± 1.2	0.037 ± 0.001
Accuracy	0.0 ± 0.0	0.0 ± 0.0	0.058 ± 0.002
Lying cost alone	0.0 ± 0.0	11.0 ± 1.5	0.219 ± 0.014
Popularity + fine	43.2 ± 5.5	100.0 ± 0.0	0.055 ± 0.004
TVC	$\mathbf{0.0 \pm 0.0}$	$\mathbf{99.0 \pm 1.2}$	$\mathbf{0.264 \pm 0.007}$
PPO			
None	99.8 ± 0.0	27.8 ± 6.4	0.086 ± 0.003
Popularity	99.8 ± 0.0	99.1 ± 0.7	0.038 ± 0.002
Accuracy	0.2 ± 0.1	0.9 ± 0.6	0.062 ± 0.003
Lying cost alone	10.5 ± 1.8	15.4 ± 2.8	0.242 ± 0.009
Popularity + fine	66.6 ± 2.8	87.4 ± 1.7	0.046 ± 0.002
TVC	$\mathbf{4.5 \pm 0.9}$	$\mathbf{97.9 \pm 0.9}$	$\mathbf{0.260 \pm 0.006}$

PPO runs. With DQN, TVC brings lying to 0.0%, and with PPO to 4.5%, with the highest throughput of the designs in each block (0.264 and 0.260, against 0.266 for REINFORCE).

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